Stealthy Computational Delay Attacks on Control Systems

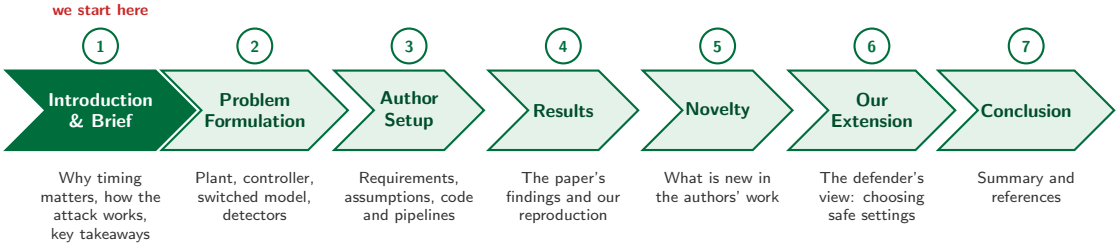
Talitha Nauta, Henrik Sandberg and Martina Maggio · ACM/IEEE ICCPS 2025

Presented by

Shubham Tyagi | **Swarali** | **Madhavi Singh** | **Abhishek Sharma** | **Srinjoy**

Embedded and Cyber-Physical Systems

Presentation Roadmap



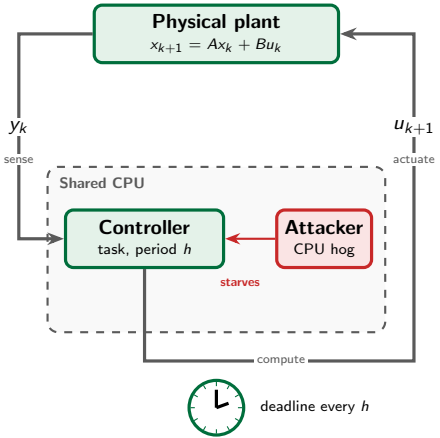
The question that runs through the whole talk

How much damage can an attacker cause **only by delaying the controller**, while staying completely **undetected**?

Source: T. Nauta, H. Sandberg and M. Maggio, "Stealthy Computational Delay Attacks on Control Systems," ICCPS 2025. DOI 10.1145/3716550.3722013

Introduction: Control Loops Run on a Clock

- A **cyber-physical system** couples computation with a physical process: power grids, factories, vehicles, medical devices.
- The controller is a **periodic task**. Every period h it reads y_k, computes u_{k+1}, and must finish before the **deadline**.
- Famous attacks such as **Stuxnet (2010)** and the **German steel mill (2014)** manipulated the *data*: sensor readings or control signals.
- Defences followed: **physics-based detectors** check whether y and u agree with the system model.



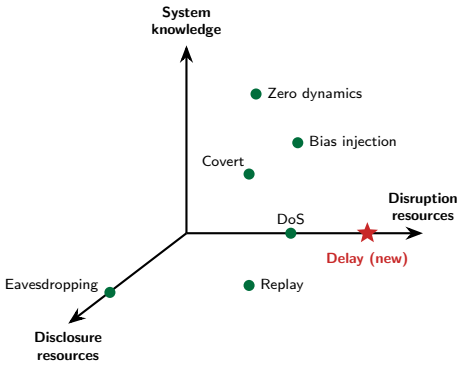
The question this paper asks

What if the attacker never touches the data, and only delays *when* the control signal is computed?

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Introduction: Where Delay Attacks Fit

Taxonomy of CPS attacks (Teixeira et al.)



The paper adds the **delay attack**: it needs only disruption resources, no access to the data.

	False data injection	**Denial of service**	**Computational delay**
What it changes	sensor or actuator values	availability	*timing* of the control job
Needs to read data?	yes	no	**no**
How it looks	model mismatch	large and obvious	like normal timing jitter
Physics detector catches it?	often	yes	**no**: y and u stay consistent

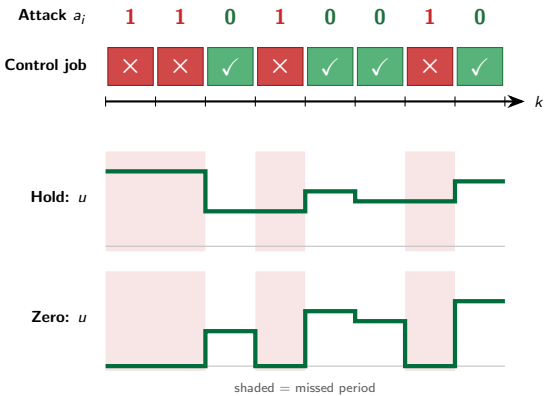
Contributions of the paper

- Extends the attack taxonomy with **computational delay attacks**.
- Finds the **worst-case stealthy attack** as an optimisation problem.
- Tests it on a **stable** and an **unstable** plant against realistic detectors.

Source: T. Nauta, H. Sandberg and M. Maggio, "Stealthy Computational Delay Attacks on Control Systems," ICCPS 2025. DOI 10.1145/3716550.3722013

Brief of the Paper: How the Attack Works

- **CPU starvation:** when a control job starts, the attacker may launch a task that holds the processor, so the job misses its deadline.
- Each period the attacker picks $a_i \in \{0, 1\}$: **0 = deadline hit**, **1 = forced miss**.



During a miss

- No new control signal is computed.
- **Hold:** keep the old u. **Zero:** set $u = 0$.
- The late job then uses the **stale** measurement $\tilde{y}_k$.

Closed-loop model

With $\psi_k = [x_k^\top \ \tilde{y}_k^\top \ z_k^\top \ u_k^\top]^\top$:

$$\psi_{k+1} = \begin{cases} \Phi_h \psi_k & \text{deadline hit} \\ \Phi_m \psi_k & \text{deadline miss} \end{cases}$$

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Brief of the Paper: What the Defender Watches

The defender cannot see a data change, so it watches the **timing** of the control task with three checks.

1. Watchdog



Shuts the plant down after $\bar{a}$ consecutive misses. In industry $\bar{a}$ is usually 3 to 10.

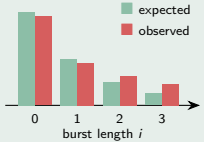
2. Sliding window test



Binomial test: alarm if the misses in the window exceed what the normal miss rate $\hat{p}_m$ allows:

$$\sum_i w_i \leq \mathfrak{B}_{1-\alpha}(n_w, \hat{p}_m)$$

3. Long-term history test



Counters m_i record bursts of i misses since start-up. A χ^2 **test** checks they follow the geometric law $E_i = \hat{p}_m^i(1-\hat{p}_m)N$.

Stealth requirement

The attack is stealthy only if **none** of these checks can reject "the system is behaving normally" at significance level α (the paper uses $\alpha = 0.01$).

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Brief of the Paper: The Optimal Stealthy Attack

Attacker's problem (MIQCP)

$$\max_{a_1,\dots,a_\ell} \quad y_\ell^\top Q\, y_\ell \qquad \text{push output away from target}$$

$$\text{s.t.} \quad a_i \in \{0,1\} \qquad \text{hit or miss}$$

$$\sum_{q=j}^{j+\bar{a}} a_q \le \bar{a} \qquad \text{no watchdog shutdown}$$

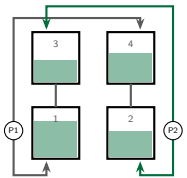
$$\psi_{i+1} = \left[(1-a_i)\Phi_h + a_i\Phi_m\right]\psi_i \qquad \text{closed-loop dynamics}$$

$$\sum_i w_i \le \mathfrak{B}_{1-\alpha}(n_w, \hat{p}_m) \qquad \text{pass binomial test}$$

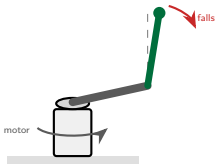
$$S_{\chi^2} \le \chi^2_{1-\alpha}(\bar{a}) \qquad \text{pass geometric test}$$

- Binary choices plus quadratic terms: a **Mixed Integer Quadratically Constrained Program**, solved with **Gurobi**.
- The answer is a schedule of misses, e.g. 10101101110100 0.

Two test plants



Quadruple tank: stable, simulated, LQG, $h = 0.1\,\text{s}$, hold



Furuta pendulum: unstable, real hardware, LQR, $h = 5\,\text{ms}$, hold and zero

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Key Takeaways

1. **Timing is an attack surface**
   No data is altered, so physics-based detectors stay silent. The attacker needs no access to sensor values.
2. **Watchdogs give little protection**
   They only help if the plant shuts down after 1 or 2 misses. From $\bar{a} \geq 3$ the attack is almost as strong.
3. **Long memory beats short memory**
   The χ^2 test on long-term counters limits the attacker far more than the sliding-window binomial test.
4. **The actuation policy matters**
   For the Furuta pendulum, *zero* is the safer choice. Under *hold*, a stealthy attack can destabilise it.
5. **Well-timed beats more**
   Optimal attacks interleave misses with hits, and around 10 periods is already enough for peak damage.

What defenders should do

1. Monitor long-term timing statistics
2. Prefer zeroing over holding on unstable plants
3. Tune the detection window to the plant
4. Treat timing anomalies as possible attacks

Bottom line

Real-time scheduling is part of security: timing must be monitored over the **long term**, not only in the last few periods.

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